Semantic Draft-and-Verify for Real-Time Activity Description on a Phone–Laptop Link

Tianchen Liu*, Juntao Hu*

***Abstract*—This project builds a phone–laptop system that describes what a person is doing with a sentence, rather than a closed activity label. The iPhone runs a small vision-language model in the browser and shows that sentence immediately. The laptop runs a CLIP encoder. A confident sentence is sent as text and mapped onto a sentence bank. An uncertain sentence is sent with the JPEG, and CLIP either keeps it or replaces it with the caption retrieved from the image. The time until the first sentence is the on-device generation time. The time until the calibrated sentence adds uplink, CLIP, and downlink, and the uplink is long only when the picture is sent. A browser–laptop prototype already separates these stages. The rest of the project measures the split on an iPhone over Wi-Fi, compares confidence-gated upload with always-upload and text-only baselines, and lets the link choose when a picture is worth sending and how large it should be. A multi-user delay model, calibrated from the measured Wi-Fi rate, studies more clients than the testbed can host.**

***Index Terms*—Human activity recognition, device–edge inference, internet of things.**

I. Motivation

An interactive or embodied system has to respond while a person is still performing the action. A closed label such as "sitting" merges sitting still, sitting and looking at a phone, and sitting and waving. Those situations call for different responses, so the system needs a sentence about the action, not one class.

The camera is on the phone. A sentence written there can appear before any server is involved. A small model next to the camera is often unsure, especially when two actions look alike. A server can check that sentence, but the check needs the picture. Sending every picture makes the user wait on the uplink. Keeping every picture on the phone leaves wrong sentences on the screen. The question is when a local sentence is safe to show, and when the picture has to cross the link so the server can correct it.

II. Literature Study

Closed activity labels are a poor fit once two actions share a posture but call for different responses. Vision-based human activity recognition assigns a clip to one name in a fixed taxonomy [1]. Benchmarks such as Kinetics train a classifier over a few hundred action classes [2]. Two actions that share a posture but differ in the object or the gesture fall into one class whenever the taxonomy has no separate name for them. Open-set action recognition keeps that taxonomy and adds a reject option. Bao *et al.* estimate uncertainty so a model can name a known action and refuse an unknown one [3]. The refused action is not described. Two variants that share a posture remain a single known class when the training set never separated them. Vision-language encoders such as CLIP place images and sentences in one space, so the description can be a full sentence retrieved by similarity, without training a new classification head [4]. That result fixes the unit of recognition. It still assumes the image is already at the encoder. It does not say whether a sentence written beside the camera, before any upload, can be kept.

A model small enough to sit next to the camera can write that sentence itself. SmolVLM shows that a vision-language model with a few hundred million parameters can generate a short description under a tight compute budget [5]. The remaining difficulty is reliability. Similar actions lie close together, and a small generator does not mark which of its sentences are safe to act on. Whether the picture must leave the device is outside the scope of making the model small.

Device–edge work already separates a fast local answer from a slower check. Speculative decoding lets a small model propose tokens and a large model accept a prefix or reject it [6], [7]. CoVSpec places a lightweight draft vision-language model on the phone and a larger target model on the edge, and it varies how often the edge verifies [8]. The edge check still consumes the visual context. Venkatesha *et al.* keep this split, add early exits in the target model, and run it on a legged robot [9]. Agreement there is a token match against a larger generator. SpotVLM uses a late answer differently: the device responds at once, and a delayed cloud description becomes context for later frames rather than a verdict on the current frame [10]. Selected frames are still uploaded. Older collaborative inference, such as Neurosurgeon, chooses a layer at which to cut the network so that compute and transmission balance [11]. The cloud part still receives activations from the picture.

Scheduling, in the cellular literature, means a base station giving time–frequency resources to users. Capozzi *et al.* survey LTE downlink schedulers that repeat this assignment according to channel quality and quality-of-service targets [12]. Multi-user edge computing moves the decision to the devices. Chen *et al.* let each user choose local execution or offload under wireless contention [13]. The offloaded object is a compute task of a given size. AWStream instead adapts how many bytes a stream sends, so that an analytics result stays accurate when bandwidth changes [14]. A base-station scheduler assumes control of resource blocks. An offloading game assumes the task either moves or stays. Neither asks, for a sentence already written beside the camera, whether the picture must follow it, nor which of several phones should be checked first.

What these lines leave open is the upload itself. A sentence can be written where the camera is, and a shared image–text

* equal contribution.

Algorithm 1 On-device sentence, server calibration

Generate s and q with SmolVLM, and display s.
$c_{\text{text}} \leftarrow \arg\max_{c \in \mathcal{C}} \cos(f_T(s), e_c)$.
if $q \geq \alpha$ **then**
  Send s only, and display c_{text}.
else
  Send s and the JPEG x.
  $c_{\text{img}} \leftarrow \arg\max_{c \in \mathcal{C}} \cos(f_I(x), e_c)$.
  if $c_{\text{img}} = c_{\text{text}}$ **then**
    Keep c_{text}, confirmed if $\cos(e_{c_{\text{img}}}, f_T(s)) \geq \tau$, else weak.
  else
    Replace the caption with c_{img}.
  end if
end if

space can test whether that sentence agrees with the picture. The test does not have to be another generator, and the picture does not have to cross the link on every frame. The open question is when the local sentence is safe to show, and when the server needs the picture in order to correct it.

III. Technical Approach

A. Pipeline

Let $\mathcal{C}$ be a bank of English activity sentences, each stored with a CLIP text embedding e_c. On the phone, SmolVLM-256M reads the frame and writes a short action phrase. The phrase is normalized to a sentence s in the same style as the bank, for example "a person is standing." Let q be the smallest token probability along that generation. The sentence s is shown at once. The picture stays on the phone unless $q < \alpha$.

If $q \geq \alpha$, the phone sends s and no image. The laptop maps the sentence onto the bank,

$$c_{\text{text}} = \arg\max_{c \in \mathcal{C}} \cos\big(f_T(s), e_c\big), \tag{1}$$

where f_T is the CLIP text tower, and the displayed sentence becomes c_{text}.

If $q < \alpha$, the phone sends s together with the JPEG x. CLIP also retrieves a caption from the image,

$$c_{\text{img}} = \arg\max_{c \in \mathcal{C}} \cos\big(f_I(x), e_c\big), \tag{2}$$

where f_I is the CLIP image tower. The phone's sentence is kept when $c_{\text{img}} = c_{\text{text}}$. It is marked confirmed when $\cos(e_{c_{\text{img}}}, f_T(s)) \geq \tau$, and weak otherwise. When the two captions differ, the displayed sentence becomes c_{img}. The prototype uses $\alpha = 0.5$ and $\tau = 0.72$. Both will be retuned on held-out clips, so that acceptance is not dominated by the high similarity that all "a person is …" sentences share. Algorithm 1 summarizes one frame.

B. Latency

Two clocks are kept separate. The first sentence does not wait for the link:

$$T_{\text{show}} = T_{\text{VLM}}. \tag{3}$$

The calibrated sentence does:

$$T_{\text{cal}} = T_{\text{VLM}} + T_{\text{up}} + T_{\text{CLIP}} + T_{\text{down}}. \tag{4}$$

T_{up} carries the sentence alone, or the sentence plus the JPEG. T_{down} is short because the server returns a caption identifier, not an image. An earlier same-machine run, with both models on the laptop CPU, put CLIP near 100 ms and SmolVLM near 5 s. Those figures are a reference for the two compute stages, not the iPhone result. T_{VLM} will be measured in the phone browser. T_{up} and T_{down} will be measured from timestamps on the phone and the laptop, not estimated from a Shannon formula.

C. Link control

The communication component does not allocate frequency resources on a base station. It chooses two quantities the testbed can actually enforce. The confidence threshold α decides which frames upload a picture: a lower α uploads fewer images. The JPEG quality and the long-edge resolution set how many bytes those frames carry. A larger image can change c_{img}, and it also increases T_{up}. We will compare a fixed $(\alpha, \text{quality})$ pair with a rule that lowers α and the JPEG size when the round-trip time or the RSSI indicates a weak link. With two phones on the same access point, the laptop calibrates the client with the lower confidence first.

If only one phone is available, the second client will be a scripted replay of recorded frames so that contention is still visible. Separately, a MATLAB model will place more than two users on a link whose rate-versus-payload curve is fitted to our Wi-Fi measurements. That simulation answers how text-only uploads and image uploads behave as the number of cameras grows. It is an extrapolation of the testbed, not a substitute for it.

IV. Implementation

A. Platform

The camera client is a web page. SmolVLM-256M-Instruct runs in that page, writes the action phrase, and records the smallest token probability. On an iPhone the weights run as 4-bit WASM. A desktop browser uses WebGPU when it is available. The page shows the on-device sentence first, then posts either the sentence or the sentence together with a JPEG. The server is FastAPI on the laptop and loads only CLIP. CLIP ViT-B/32 maps the sentence onto the bank and, when an image arrives, retrieves c_{img}. The caption bank is a JSON file of roughly twenty-five compositional sentences, including contrasts such as "a person is sitting still" and "a person is sitting and looking at a phone."

The iPhone client is the same page in Safari, opened over the laptop's LAN address. Live camera access on iOS requires HTTPS, so the laptop will present a locally trusted certificate. Still images from the camera roll can be used to debug the path before the certificate is in place. The front camera is used when the phone is worn or held for a self-view; the back camera is used when the phone watches another person.

B. What is already running

The browser–laptop loop is implemented. A frame returns the on-device phrase, its minimum token probability, and T_{VLM}. The server returns whether an image was uploaded, the CLIP caption when an image was sent, the agreement, and one of "confirmed," "corrected," or "weak." Uplink and downlink are still small when the browser and the server share a machine, which is why the iPhone hop has to be measured separately. The earlier CPU figures, about 100 ms for CLIP and about 5 s for SmolVLM, are kept only as a reference for those two stages.

C. Evaluation

We will record short sessions of one person performing the behaviors named in the bank, indoors, on the same Wi-Fi network used for the demo. For each session we report:

- time to the first sentence and time to the calibrated sentence, split into on-device generation, uplink, CLIP, and downlink;
- fraction of frames that upload an image, and the fractions confirmed, corrected, and weak;
- whether a viewer agrees that the final sentence describes the frame, on a small labeled subset;
- bytes uploaded per minute.

The three systems under comparison are CLIP on every uploaded frame, the on-device sentence with text-only calibration, and confidence-gated upload at a chosen α. The link-control experiment repeats confidence-gated upload at two or three operating points of α and JPEG size, including the RTT-based rule. The MATLAB study reports calibration delay versus the number of users for round-robin service and for lower-confidence-first service, using the fitted rate curve.

D. Roles

Both members integrate the demo and the report. One member owns the caption bank, the acceptance rule, the on-device vision-language client, and the CLIP server. The other owns timestamped delay logs, the iPhone network path, the choice of α and image size, and the calibrated multi-user simulation. The acceptance rule is the information-processing core. The delay breakdown and the upload rule are the system core required by the course.

V. Project Timelines

The browser–laptop loop is already in place. The remaining work ends at the final presentation on 20 November 2026.

TABLE I
SCHEDULE

Dates	Work
23 Sep.–3 Oct.	iPhone capture on Wi-Fi, HTTPS, and split delay logs
4–17 Oct.	Caption bank, thresholds α and τ, and the three baselines
18–31 Oct.	Link rule for upload and image size; two clients on one access point
1–12 Nov.	Fit the rate curve and run the multi-user MATLAB comparison
13–20 Nov.	Demo, report, and presentation

References

[1] Y. Kong and Y. Fu, "Human action recognition and prediction: A survey," *International Journal of Computer Vision*, vol. 130, no. 5, pp. 1366–1401, 2022.

[2] J. Carreira and A. Zisserman, "Quo vadis, action recognition? a new model and the Kinetics dataset," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 6299–6308.

[3] W. Bao, Q. Yu, and Y. Kong, "Evidential deep learning for open set action recognition," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 13 349–13 358.

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *International Conference on Machine Learning*, 2021.

[5] A. Marafioti, O. Zohar, M. Farré *et al.*, "SmolVLM: Redefining small and efficient multimodal models," *arXiv preprint arXiv:2504.05299*, 2025.

[6] Y. Leviathan, M. Kalman, and Y. Matias, "Fast inference from transformers via speculative decoding," in *International Conference on Machine Learning*, 2023.

[7] C. Chen, S. Borgeaud, G. Irving, J.-B. Lespiau, L. Sifre, and J. Jumper, "Accelerating large language model decoding with speculative sampling," *arXiv preprint arXiv:2302.01318*, 2023.

[8] Y. Jia, S. Tang, and Q. Yang, "CoVSpec: Efficient device–edge co-inference for vision-language models via speculative decoding," *arXiv preprint arXiv:2605.02218*, 2026.

[9] Y. Venkatesha, S. Kundu, and P. Panda, "Fast and cost-effective speculative edge-cloud decoding with early exits," *arXiv preprint arXiv:2505.21594*, 2025.

[10] C. Qian, X. Yu, Z. Huang, D. Li, Q. Ma, F. Dang, X. Ding, G. Shang, and Z. Yang, "SpotVLM: Cloud-edge collaborative real-time VLM based on context transfer," *arXiv preprint arXiv:2508.12638*, 2025.

[11] Y. Kang, J. Hauswald, C. Gao, A. Rovinski, T. Mudge, J. Mars, and L. Tang, "Neurosurgeon: Collaborative intelligence between the cloud and mobile edge," in *Proceedings of the Twenty-Second International Conference on Architectural Support for Programming Languages and Operating Systems*, 2017, pp. 615–629.

[12] F. Capozzi, G. Piro, L. A. Grieco, G. Boggia, and P. Camarda, "Downlink packet scheduling in LTE cellular networks: Key design issues and a survey," *IEEE Communications Surveys & Tutorials*, vol. 15, no. 2, pp. 678–700, 2013.

[13] X. Chen, L. Jiao, W. Li, and X. Fu, "Efficient multi-user computation offloading for mobile-edge cloud computing," *IEEE/ACM Transactions on Networking*, vol. 24, no. 5, pp. 2795–2808, 2016.

[14] B. Zhang, X. Jin, S. Ratnasamy, J. Wawrzynek, and E. A. Lee, "AWStream: Adaptive wide-area streaming analytics," in *Proceedings of the ACM SIGCOMM Conference*, 2018.